

Zachary Nason

Phone: 19136452840
Email: znason2@huskers.unl.edu
zach.nason96@gmail.com
Website: <https://zach-nason.github.io/>

EDUCATION

B.S., Mathematics 2020
University of Kansas, Lawrence, KS

M.S., Mathematics 2023
University of Nebraska-Lincoln, Lincoln, NE

Ph.D., Mathematics Expected May 2027
University of Nebraska-Lincoln, Lincoln, NE
Advisor: Thomas Marley

POSITIONS HELD

Graduate Teaching Assistant August 2021 - Present
Mathematics Department, University of Nebraska-Lincoln

Research Training Group in the Mathematical Sciences (RTG) Graduate Trainee Fall 2024
Mathematics Department, University of Nebraska-Lincoln

AWARDS

Emeritus Faculty Fellowship - Lloyd Jackson Award 2025
This award is given by the UNL Mathematics Department to graduate students to recognize notable scholarship and to promote future mathematical research. This award is named after UNL emeritus professor Lloyd Jackson.

Outstanding Qualifying Exam 2022
This award is given by the UNL Mathematics Department to two graduate students who have achieved the highest scores on the Ph.D. qualifying exams for that year.

PUBLICATIONS/PREPRINTS

- [1] Lars Winther Christensen, Antonia Kekkou, Justin Lyle, Zachary Nason, and Andrew J. Soto Levins, *G-levels of perfect complexes and a Bass formula for levels*, Accessible at <https://arxiv.org/abs/2507.10486>. Currently under review at Proceedings of the American Mathematical Society, April 2026.
- [2] Paulo Martins, Victor D. Mendoza Rubio, and Zachary Nason, *Finiteness of complete intersection dimensions of $R\text{Hom}$ complexes and Ext modules*, Accessible at <https://arxiv.org/abs/2601.07811>. Currently under review at Journal of Pure and Applied Algebra, January 2026.
- [3] Zachary Nason, *Maximal Cohen-Macaulay DG-complexes*, Journal of Pure and Applied Algebra **230** (2026), no. 2, 108193.
- [4] Zachary Nason, Andrew J. Soto Levins, and Ryan Watson, *Quasi-Gorenstein morphisms of commutative local dg-algebras*, Accessible at <https://arxiv.org/abs/2605.02029>. Currently under review at Bulletin of the London Mathematical Society, May 2026.

TALKS AND PRESENTATIONS

NON-UNL PRESENTATIONS

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|---|------------------|
| 4. Maximal Cohen-Macaulay DG-complexes (50 min)
<i>University of Texas-Rio Grande Valley, Algebra Seminar</i> | April 13, 2026 |
| 3. Level inequalities for complexes (50 min)
<i>Texas Tech University, Algebra and Number Theory Seminar</i> | March 4, 2026 |
| 2. Level inequalities for complexes (50 min)
<i>Syracuse University, Algebra Seminar</i> | January 30, 2026 |
| 1. Maximal Cohen-Macaulay DG-complexes and a derived
Improved New Intersection Theorem (20 min)
<i>University of Kansas, AMS Spring Central Sectional 2025
Special Session on Topics in Commutative Algebra</i> | March 29, 2025 |

UNL PRESENTATIONS

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| 11. Koszul property for points in projective space (50 min)
<i>University of Nebraska-Lincoln, RTG Learning Seminar: Koszul Algebras</i> | April 6, 2026 |
| 10. Level inequalities for complexes (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Seminar</i> | December 3, 2025 |
| 9. Journeying towards Jensen's theorem (3 talks, 50 min each)
<i>University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar</i> | July 21, 23, and 25, 2025 |
| 8. Maximal Cohen-Macaulay DG-modules (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Seminar</i> | November 20, 2024 |
| 7. Maximal Cohen-Macaulay complexes and an extension to
differential graded rings (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | September 25, 2024 |
| 6. Constructing the derived category (3 talks, 50 min each)
<i>University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar</i> | July 29, 31, and August 2, 2024 |
| 5. Cohen-Macaulay differential graded algebras (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | May 1, 2024 |
| 4. Gorenstein differential graded algebras (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | October 19, 2023 |
| 3. Characterizing Gorenstein local rings using the Frobenius map (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | April 19, 2023 |
| 2. Symbolic powers and monomial ideals (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | November 16, 2022 |
| 1. A counterexample concerning the existence of height one
prime ideals in the intersection of height two prime ideals (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | February 9, 2022 |

CONFERENCES ATTENDED

- A Celebration of Golod's Mathematical Legacy
University of Warwick April 2026
- CA+ 2026
Iowa State University March 2026
- SLAM 2026
The University of Texas at Arlington March 2026
- KUMUNU
University of Missouri October 2025
- AMS Fall 2025 Southeastern Sectional Meeting
Tulane University October 2025
- Summer CAMP
University of Nebraska-Lincoln August 2025
- Modules and Rings: Recent Developments in Commutative Algebra
University of Genova June 2025
- URiCA
University of Nebraska-Lincoln May 2025
- RayFest
University of Nebraska-Lincoln April 2025
- AMS Spring 2025 Central Sectional Meeting
University of Kansas March 2025
- Commutative Algebra in Mixed Characteristic
University of Nebraska-Lincoln March 2025
- CA+ 2025
Iowa State University March 2025
- KUMUNU
University of Missouri September 2024
- URiCA
University of Nebraska-Lincoln May 2024
- KUMUNU
University of Missouri October 2023
- AMS Fall 2023 Central Sectional Meeting
Creighton University September 2023
- BrianFest
University of Nebraska-Lincoln August 2023
- URiCA
University of Nebraska-Lincoln May 2023
- Morgantown Algebra Days
West Virginia University April 2023
- Math 125
University of Nebraska-Lincoln April 2023
- KUMUNU
University of Nebraska-Lincoln October 2022

TEACHING

Recitation Instructor MATH 221: <i>Differential Equations</i>	Spring 2026 (2 sections)
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2025
Instructor of Record MATH 107WHT: <i>Calculus II - William H. Thompson Scholars Section</i>	Spring 2025
Instructor of Record MATH 208: <i>Calculus III</i>	Summer 2024
Instructor of Record MATH 104: <i>Applied Calculus</i>	Spring 2024
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2023
Grading Assignment MATH 825: <i>Mathematical Analysis I</i>	Fall 2023
Instructor of Record MATH 107: <i>Calculus II</i>	Summer 2023
Algebra Qualifying Exam Review Session Organizer Organized and led the review session for UNL's algebra qualifying exam.	Summer 2023
Recitation Instructor MATH 208: <i>Calculus III</i>	Spring 2023 (3 sections)
Instructor of Record MATH 101: <i>College Algebra</i>	Fall 2022
Recitation Instructor MATH 107: <i>Calculus II</i>	Spring 2022 (2 sections), Summer 2022 (1 section)
Recitation Instructor MATH 106: <i>Calculus I</i>	Fall 2021
Tutoring Drop-in math tutoring center for UNL first-year undergraduates. Tutored for two hours per week, one semester per year.	Fall 2021-Present

SERVICE

Commutative Algebra Reading Seminar Co-Organizer <i>University of Nebraska-Lincoln</i> Organized and led a weekly graduate student seminar in commutative algebra along with two other graduate students.	2025-Present
UNL DRP Mentor <i>University of Nebraska-Lincoln</i> Mentored an advanced undergraduate student in basic commutative algebra and planar algebraic geometry, with the eventual goal of proving Bezout's Theorem for projective curves in projective 2-space	2024-2025
UNL Graduate Mentor <i>University of Nebraska-Lincoln</i> Mentored a beginning mathematics graduate student during his first year of graduate school. Advised and helped the first-year graduate students with navigating academic life and with adjusting to Lincoln.	2023-2024

UNL Math Day Volunteer

2021-Present

University of Nebraska-Lincoln

The mathematics department at the University of Nebraska-Lincoln holds an annual math day for high school students in Nebraska. Students compete in a math bowl and participate in mathematical activities and puzzle-solving.

COMPUTER SKILLS

Languages: Java, Python, C++

Software: $\text{\LaTeX}$, Macaulay2, MATLAB, Microsoft Office, Google Workspace